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ICS 2203 WEB APPLICATION DEVELOPMENT 1

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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LESSON 1

The Internet and Networks

Learning outcomes

Upon completing this topic, you should be able to:

1. Define the Internet and its components.
2. Identify Various Network types.
3. Relate to the Dynamic nature of the Internet.

1.1. Introduction

Previously you may recall learning the following in the Computer Networks Class in stage one (1) semester two (2)

- Layered communication protocols



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- Open Systems Interconnection (OSI) reference model
- Computer networking standards and standardization process

These are going to be vital in this class so if you have forgotten them you may want to remind yourself about them. We will begin this lesson with some definitions of key terms. The first one being “what is the Internet?”. If you did a bit of research online, the term internet would give you results such as a global interconnection of computer networks. Based on the History of the Internet this is partly correct. The Internet has evolved quite a bit from the time of its inception. Initially as a simple network of a few computers in a room. This later in cooperated telephone networks to connect to a gigantic global Digital Pipeline interconnecting different components that share resources.



1.2. The Internet and Computer Networks

The most common definition of the internet is; the internet is a network of Computers. A good starting point to explore this therefore is in computer networks.

1.3. Motivation for Networks

Networks are used to facilitate communicate between individual devices or people. The other motivation for networks is the sharing of resources; network makes it possible to share data and resources.

- Data: share files, share a database.
- Resources: on a local network, it is quite common to share a printer (for example), amongst several computers.



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The Internet is a heterogeneous network (network of networks), currently comprising of networks such as wide area cellular networks (GPRS), and gig speed networks such as for digital television, telephone networks requiring fiber or satellite technologies.

1.4. Network Types

Computer networks are groups of computers interconnected via physical or non-physical medium. They are designed to allow computers to communicate with each other and share resources.

Networks in general may be classified based on the following:

- Scale
- Connection method
- Network Architectures (or functional relationship)



- Network topology

1.5. Types of Computer networks include

- Metropolitan area network
- Personal area network
- Local area network
- Campus area network
- Wide area network
- Virtual private network
- Global area network

1.6. Benefits of the internet

- Fast seamless communication saving on time



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- Shared resources globally- limitless access to information and resources globally
- Much cheaper form of communication compared to other forms i.e. telephony, fax or air mail.
- Security and privacy to some extent
- Global reach spanning vast geographical space.

Example . The internet is a System of Systems. Discuss

Solution: Points-Network of networks (MAN s, WANs, LANs etc), Different platforms of computers, Different mediums, Many standards of communication, Many resources to be shared, Dynamic uncontrolled growth





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Learning Activities

Revision Questions

EXERCISE 1. ✍ Discuss five benefits that are associated with the Global Internet?

EXERCISE 2. ✍ The following are some of the ills associated with the internet. Discuss these

- i. Theft of Personal Information
- ii. Spamming
- iii. Virus Threat .

EXERCISE 3. ✍ Define the following terms in relation to the Internet and the World Wide Web.

1. Protocol:
2. VPN:



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3. CGI:
4. Html:
5. Fibre Optic

EXERCISE 4.  Describe the significance of the following terms to the internet

1. Browser:
2. Server:
3. Domain:
4. IP:
5. DNS:

References and Additional Reading Materials

- George Beekman,(2008) Tomorrow's Technology and You, Complete, 9/E,Oregon State University Ben Beekman, ISBN-



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- Chuck Musciano and Bill Kennedy(2000)HTML & XHTML: The Definitive Guide, 4th Edition, Oreilly, ISBN 0-596-00026-X
- Craig Grannell(2007 The Essential Guide to CSS and HTML Web Design, Springer, ISBN-13 (pbk): 978-1-59059-907-5

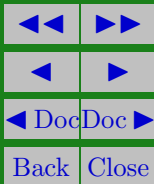


Solutions to Exercises

Exercise 2. i. Theft of Personal Information If you use the Internet for online banking, social networking or other services, you may risk a theft to your personal information such as name, address, credit card number etc. Unscrupulous people can access this information through unsecured connections or by planting software and then use your personal details for their benefit. Needless to say, this may land you in serious trouble. ii. Spamming Spamming refers to sending unwanted e-mails in bulk, which provide no purpose and needlessly obstruct the entire system. Such illegal activities can be very frustrating for you as it makes your Internet slower and less reliable. iii. Virus Threat Internet users are often plagued by virus attacks on their systems. Virus programs are



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Exercise 3. i. Protocol: these facilitate communication with other computers on the network following OSI Networking Model i.e. Application layer onwards. ii. VPN: Virtual private network is a computer network in which some of the links between nodes are carried by open connections or virtual circuits in some larger network (e.g., the Internet) iii. CGI: Common Gateway Interface CGI is the part of the Web server that can communicate with other programs running on the server iv. Html: Hypertext Mark up Language that allows for creation of web pages through insertion of special tags. v. Fibre Optic: Physical medium for creation of ultra fast networks supporting great band widths

Exercise 3



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Exercise 4. i. Browser: Clients software that allows a computer user to request for a resource from a server ii. Server: this may be a physical machine or service that is responsible to acting on any request by clients iii. Domain: this represents a network segments that is administered over by a sever. iv. IP: The Ip identifies machines and other resources on the network v. DNS: Domain Name Service is responsible for interpreting domain names to their equivalent numerical IP addresses

Exercise 4